

A finite-cell counterexample to higher-order Minkowski increments

A negative answer to Question 3.10 of arXiv:2206.01565

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12 August 2026

Status. High-confidence candidate full counterexample, pending human review. Question 3.10 is false at order $m = 5$ in every dimension $n \geq 2$.

The source question

For compact $A, B \subset \mathbb{R}^n$ with $0 \in B$, put $\Delta_B(A) = (A + B) \setminus A$. The source proves its proposed inequality for $m = 2$ and asks whether, for every m , every compact B_0 , and compact $B_1, \dots, B_m$ containing zero,

$$|\Delta_{B_1} \cdots \Delta_{B_m}(B_0)| \geq \sum_{S \subseteq [m]} (-1)^{m-|S|} |B_0 + \sum_{i \in S} B_i|. \quad (\text{Q})$$

this can be seen, for example, in $\mathbb{R}^2$ by taking A to be a square, B to be a segment, and C to be a Euclidean ball.

It is natural to ask if the higher-order analog of this observation remains true.

Question 3.10. *Let $m \in \mathbb{N}$ and $B_1, \dots, B_m \subset \mathbb{R}^n$ be compact sets containing the origin. Then, for any compact $B_0 \subset \mathbb{R}^n$, is it true that*

$$|\Delta_{B_1} \cdots \Delta_{B_m}(B_0)| \geq \sum_{s \subseteq [m]} (-1)^{m-|s|} \left| B_0 + \sum_{i \in s} B_i \right|?$$

The inequality (23) gives a positive answer to the above question in the case $m = 2$. We also observe that if $B_0, B_1, \dots, B_m$ are convex, then the right hand side is non-negative thanks to Theorem 3.6. We note that it was observed in [20], by considering $A = \{0, 1\}$ and $B = C = [0, 1]$ in $\mathbb{R}^1$, that the volume of Minkowski sums cannot be supermodular (even in dimension 1) if the convexity assumption on the set A is removed. Nonetheless [20] observed that if $A, B, C \subset \mathbb{R}$ are compact, then

$$|A + B + C| + |\text{conv}(A)| \geq |A + B| + |A + C|;$$

it is unknown if this extends to higher dimension. In particular, we do not know if the following conjecture is true for $n \geq 2$.

Conjecture 3.11. *For any convex body A and any compact sets B and C in $\mathbb{R}^n$,*

$$|A + B + C| + |A| > |A + B| + |A + C|.$$

The exact page is bundled as `source_excerpt_question_3_10_page_13.pdf`.

Theorem 1. *For every $n \geq 2$ and every $0 < \varepsilon < 1$, there are a compact B_0 and finite compact sets $B_1, \dots, B_5 \subset \mathbb{R}^n$, each containing zero, for which the left side of (Q) is $4\varepsilon^n$ and the right side is $6\varepsilon^n$. Hence (Q) is false.*

The discrete configuration

Work first in $\mathbb{Z}^2$. Define

$$\begin{aligned} E_0 &= \{(-1, 0), (1, -2), (-1, -2)\}, \\ P_1 &= \{(-2, 0), (-2, -1), (0, 0)\}, & P_2 &= \{(1, 0), (0, 0)\}, \\ P_3 &= \{(1, -1), (0, 0)\}, & P_4 &= \{(0, -1), (0, 0)\}, \\ P_5 &= \{(2, -1), (-1, -1), (0, 0)\}. \end{aligned}$$

For finite lattice sets put $\delta_P(E) = (E + P) \setminus E$ and recursively $E_i = \delta_{P_i}(E_{i-1})$. Direct set arithmetic gives

i	$ E_i $	E_i
0	3	$\{(-1, 0), (1, -2), (-1, -2)\}$
1	5	$\{(-3, 0), (-3, -1), (-3, -2), (-3, -3), (-1, -3)\}$
2	5	$\{(-2, 0), (-2, -1), (-2, -2), (-2, -3), (0, -3)\}$
3	5	$\{(-1, -1), (-1, -2), (-1, -3), (-1, -4), (1, -4)\}$
4	2	$\{(-1, -5), (1, -5)\}$
5	4	$\{(-2, -6), (0, -6), (1, -6), (3, -6)\}$

Thus the iterated discrete increment has cardinality four.

For a mask $M \in \{0, 1\}^5$, let $P_M = \sum_{i:M_i=1} P_i$. In binary-mask order from 00000 to 11111, the 32 values of $|E_0 + P_M|$ are

$$(3, 8, 6, 16, 6, 16, 12, 26, 6, 12, 12, 24, 12, 24, 18, 33, \\ 9, 22, 17, 34, 16, 35, 26, 45, 16, 30, 26, 43, 27, 44, 35, 55).$$

Grouping by Hamming weight $r = |M|$ gives respectively

$$(N_0, N_1, N_2, N_3, N_4, N_5) = (3, 35, 151, 270, 200, 55).$$

Therefore the fifth alternating difference is

$$-N_0 + N_1 - N_2 + N_3 - N_4 + N_5 = -3 + 35 - 151 + 270 - 200 + 55 = 6. \quad (1)$$

Exact lift to compact subsets of $\mathbb{R}^n$

Embed every displayed lattice point in the first two coordinates of $\mathbb{Z}^n$ and let $Q = [0, \varepsilon]^n$. For finite $E \subset \mathbb{Z}^2$ write

$$\Phi(E) = (E \times \{0\}^{n-2}) + Q.$$

Because $0 < \varepsilon < 1$, distinct cubes in $\Phi(E)$ have positive separation. Consequently, for all finite lattice sets E, F, P ,

$$\Phi(E) + P = \Phi(E + P), \quad \Phi(E) \setminus \Phi(F) = \Phi(E \setminus F), \quad |\Phi(E)| = |E|\varepsilon^n, \quad (2)$$

where P is embedded in the same coordinate plane. The middle equality is used only for unions of cells from this common lattice family.

Take $B_0 = \Phi(E_0)$ and take the five increment sets to be the finite embedded sets P_i . Every P_i contains zero. From (2), if $D_0 = B_0$ and $D_i = \Delta_{P_i}(D_{i-1})$, then $D_i = \Phi(E_i)$. Hence

$$|\Delta_{P_5} \Delta_{P_4} \Delta_{P_3} \Delta_{P_2} \Delta_{P_1}(B_0)| = |D_5| = 4\varepsilon^n. \quad (3)$$

The right side of (Q) is symmetric in its five increment sets, and (2)–(1) show that it equals $6\varepsilon^n$. Thus, taking the source labels $(B_1, \dots, B_5) = (P_5, \dots, P_1)$ if its composition is read outermost first, (Q) asserts $4\varepsilon^n \geq 6\varepsilon^n$, a contradiction. This proves Theorem 1. $\square$

Proof intuition and scope

Separated lattice cubes turn Minkowski addition by a finite lattice set into ordinary finite sumset formation, while Δ becomes literal set difference of cell indices. The question therefore contains a purely finite combinatorial assertion. At the fifth iteration the boundary operation collapses the configuration to four cells, but the fifth alternating sum of all Minkowski expansions still counts six cells' worth of volume.

The construction disproves the universal question for $m = 5$ and all $n \geq 2$. It makes no claim about $m = 3$ or $m = 4$, about a convex B_0 , or about the paper's separate Conjecture 3.11. Exact-title, exact-question, and official-arXiv searches through 12 August 2026 found no later explicit answer; this is a bounded novelty audit, not a priority claim.

Verification and reference

The bundled `verifier.py` recomputes all six stages, all 32 subset cardinalities, the grouped totals, and the strict deficit using only exact integer set arithmetic.

[1] M. Fradelizi, M. Madiman, and A. Zvavitch, *Sumset estimates in convex geometry*, arXiv:2206.01565, Question 3.10, p. 13.